

ALBARTVILLE, AL, USA

WMO: 720376

Lat: 34.229N

Lon: 86.256W

Elev: 315

StdP: 97.60

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.0	(c) -4.1	(d) -14.0	(e) 1.2	(f) -1.8	(g) -12.2	(h) 1.4	(i) -0.8	(j) 10.0	(k) 4.2	(l) 8.8	(m) 6.2	(n) 3.4	(o) 350	(p) 0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.1	22.8	32.8	23.3	32.2	23.2	26.8	30.0	26.2	29.4	25.3	28.4	2.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.3	28.6	25.2	21.1	27.8	24.8	20.6	27.5	86.4	31.0	82.9	29.0	79.9	28.5	36.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.1	7.2	5.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-9.2	36.0	2.8	1.9	-11.2	37.4	-12.9	38.5	-14.4	39.6	-16.4	40.9		
			-10.5	28.4	2.4	2.9	-12.1	30.4	-13.5	32.1	-14.8	33.7	-16.5	35.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.1	7.5	12.5	16.7	20.8	24.9	26.0	25.9	23.2	16.9	10.7	8.0
	DBStd	8.47	5.72	5.41	5.47	4.27	3.59	2.14	1.94	2.32	3.08	4.51	4.55	5.25
	HDD10.0	475	173	104	40	5	0	0	0	0	0	4	45	104
	HDD18.3	1645	411	304	191	80	20	0	0	0	4	82	231	322
	CDD10.0	2868	21	34	116	205	336	448	495	492	397	217	66	42
	CDD18.3	996	0	1	9	30	98	198	237	233	151	37	2	1
	CDH23.3	8619	0	1	53	219	757	1824	2143	2139	1201	273	8	1
	CDH26.7	3284	0	0	4	29	207	746	892	920	428	58	1	0
Wind	WSAvg	2.5	3.1	3.2	3.2	2.9	2.2	1.9	1.7	1.7	1.9	2.4	2.7	2.9
Precipitation	PrecAvg	1403	143	133	144	109	113	113	109	99	99	82	115	130
	PrecMax	1757	231	262	242	194	264	257	194	230	218	236	262	255
	PrecMin	890	54	59	32	59	8	34	38	37	3	1	60	62
	PrecStd	203	44	50	47	36	64	60	49	43	58	62	48	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	22.4	27.3	29.1	32.1	35.1	36.0	37.2	33.7	31.0	24.2	22.2
		MCWB	15.3	16.9	17.1	18.0	21.1	22.2	22.6	22.4	21.1	20.5	17.3	19.6
	2%	DB	17.5	20.1	25.0	27.4	30.2	32.9	33.7	34.9	32.4	27.8	22.1	20.0
		MCWB	14.6	15.7	16.5	18.3	21.0	22.7	24.0	23.1	21.8	19.8	15.6	17.7
	5%	DB	16.1	17.8	22.5	26.1	28.9	32.3	32.4	32.7	31.1	26.3	20.1	17.8
		MCWB	14.0	13.4	15.9	17.9	20.8	22.7	24.2	23.6	22.4	18.8	15.6	16.1
	10%	DB	13.9	16.2	21.0	24.0	27.5	31.1	31.2	31.3	29.1	24.8	17.9	16.3
		MCWB	11.9	12.2	14.8	17.2	20.3	22.7	24.0	23.4	22.1	18.2	14.5	14.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	18.4	19.7	22.4	24.4	26.9	27.8	27.2	26.1	24.6	21.1	20.8
		MCDB	17.9	21.0	22.3	25.9	27.4	30.7	31.1	29.9	29.3	27.0	22.0	21.3
	2%	WB	16.0	17.3	18.4	20.8	23.3	26.1	27.0	26.5	25.2	22.8	18.9	18.3
		MCDB	16.6	18.8	21.8	23.9	26.9	29.9	30.3	29.5	28.6	25.3	20.2	19.0
	5%	WB	14.7	15.6	17.4	19.4	22.4	25.1	26.5	25.7	24.4	21.7	17.3	17.0
		MCDB	16.1	16.9	20.1	23.6	26.4	28.8	29.7	28.8	28.1	24.6	18.1	17.6
	10%	WB	12.4	13.2	16.3	18.4	21.6	24.1	25.4	25.0	23.4	20.2	15.7	14.9
		MCDB	13.8	15.1	19.5	22.5	25.7	28.3	28.1	28.4	26.7	23.2	17.5	16.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.4	10.3	11.5	10.7	10.8	10.0	10.2	10.4	11.0	10.8	8.5
		MCDBR	10.1	11.6	12.0	12.3	12.3	12.5	11.6	12.1	12.2	12.6	12.0	9.7
		MCWBR	7.7	7.8	6.4	5.5	4.8	4.2	4.2	3.9	4.4	6.0	7.3	8.0
	5% WB	MCDBR	8.5	9.2	10.0	10.5	9.9	10.3	10.1	9.8	9.9	10.0	9.0	9.1
		MCWBR	7.6	7.9	6.4	5.9	5.1	5.5	5.7	5.1	5.1	6.0	7.5	8.6
Clear-Sky Solar Irradiance	taub	0.311	0.315	0.355	0.385	0.439	0.471	0.517	0.517	0.421	0.369	0.334	0.313	
	taud	2.518	2.503	2.411	2.313	2.205	2.149	2.042	2.028	2.288	2.423	2.462	2.528	
	Ebn at Noon	902	936	918	902	852	821	781	773	839	862	864	878	
	Edh at Noon	83	93	111	128	145	153	169	169	124	100	87	78	
All-Sky Solar Radiation	RadAvg	2.52	3.04	4.19	5.43	5.94	6.19	5.92	5.50	4.77	3.92	2.87	2.15	
	RadStd	0.19	0.39	0.39	0.37	0.47	0.48	0.36	0.29	0.43	0.50	0.31	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+113

Nomenclature: See separate page